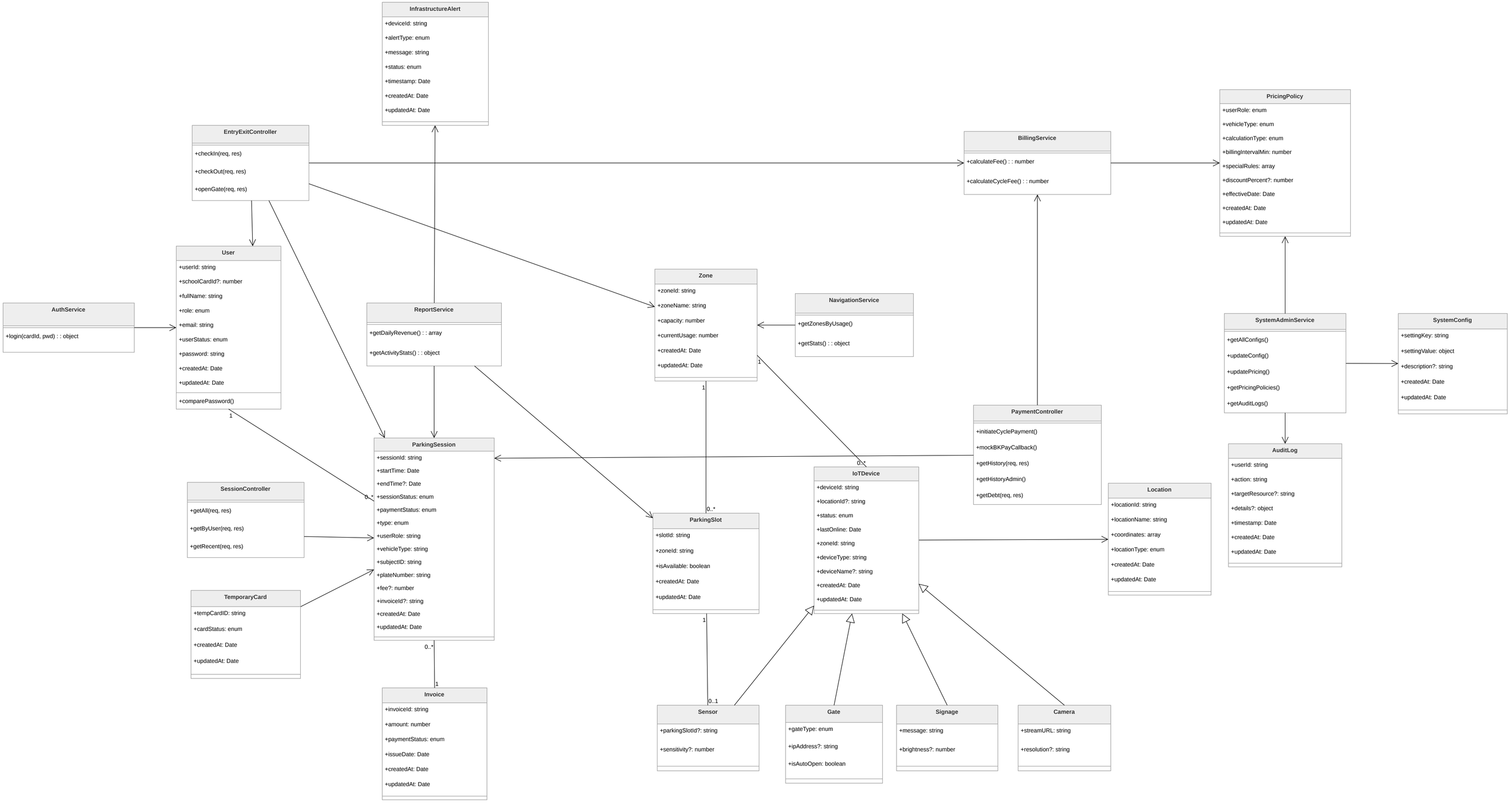
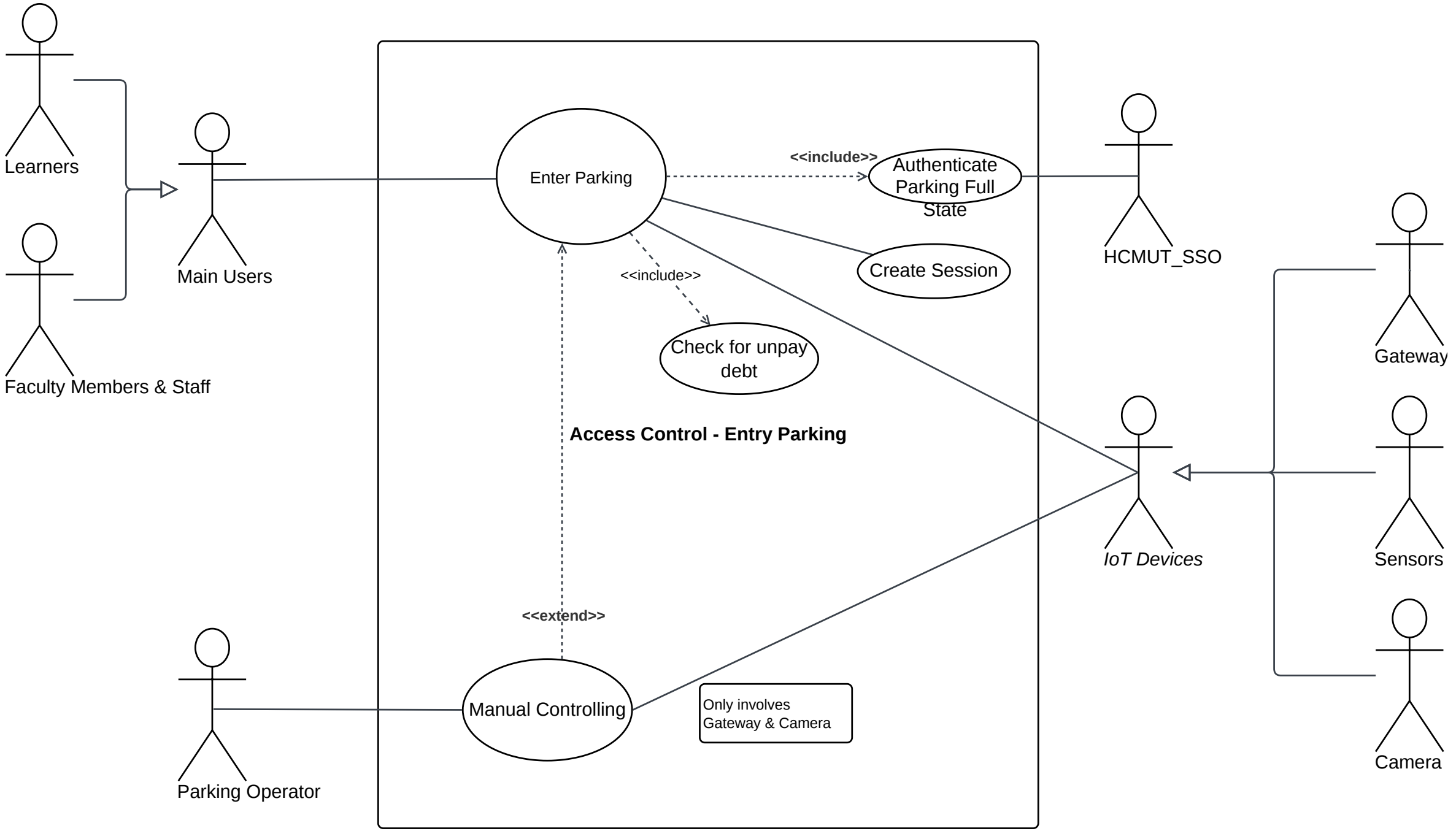
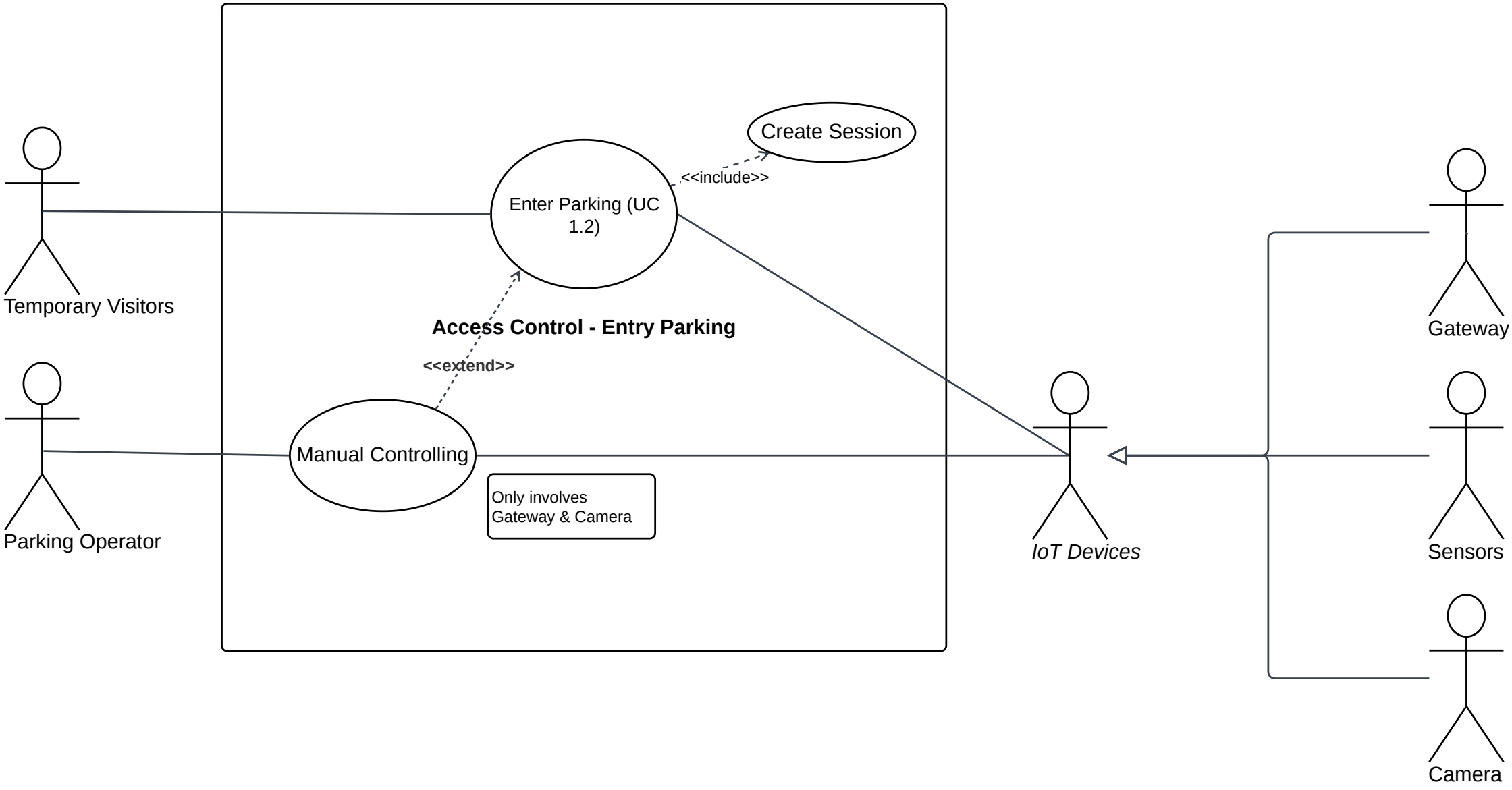
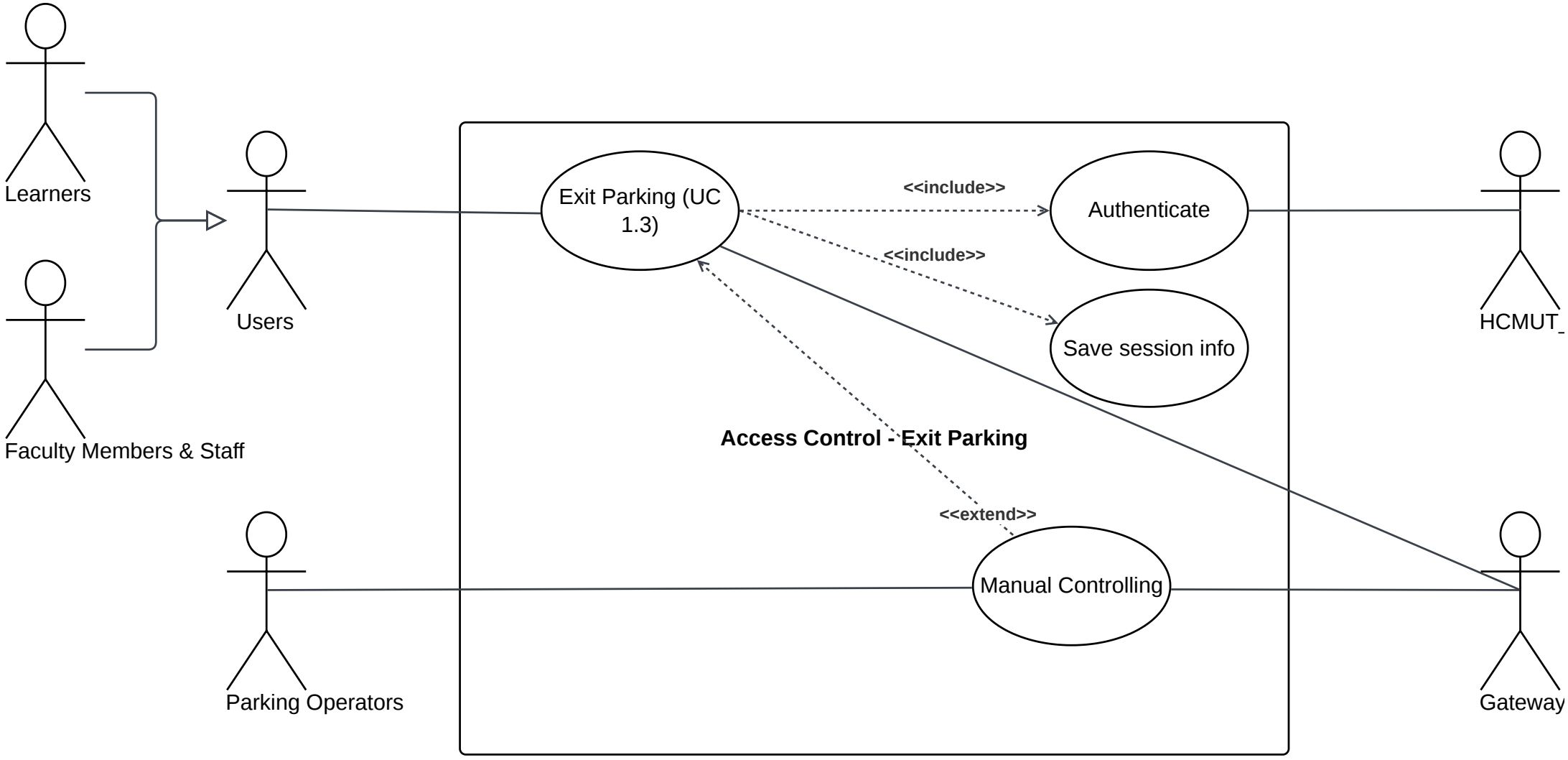
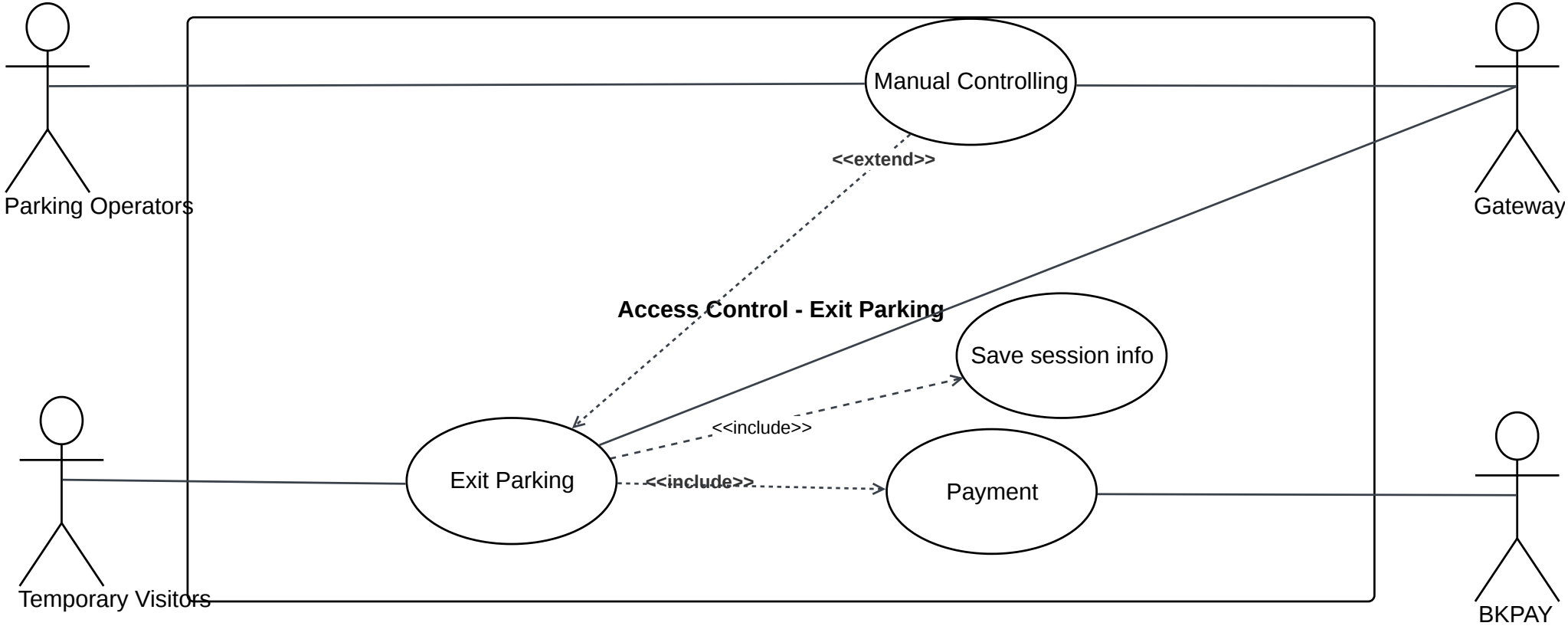
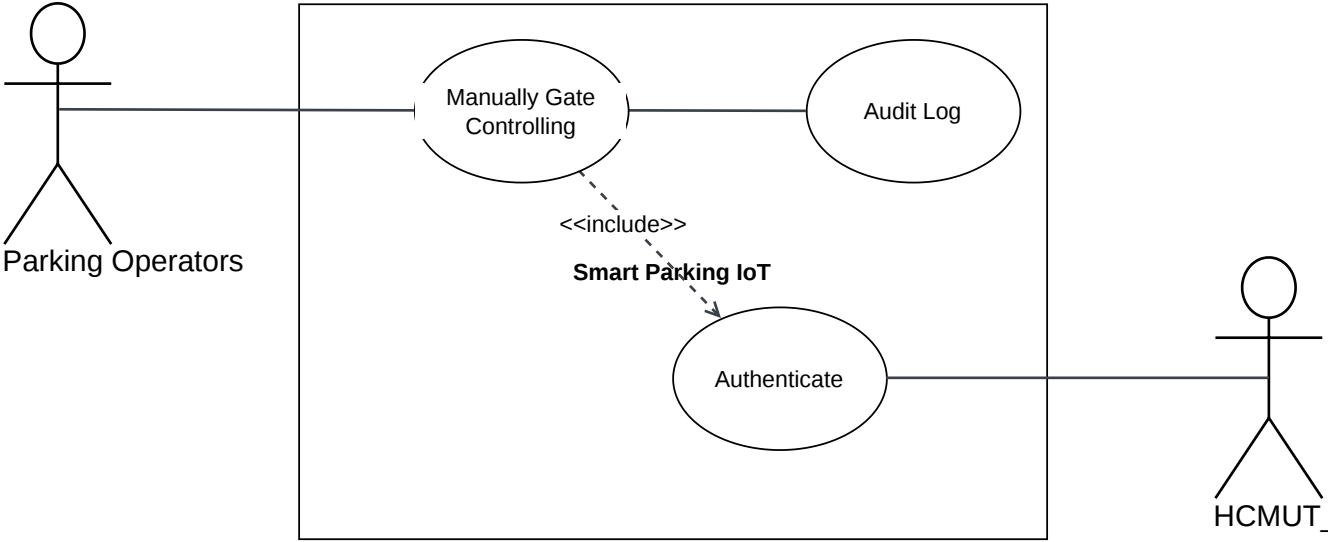

InfrastructureAlert
+deviceId: string
+alertType: enum
+message: string
+status: enum
+timestamp: Date
+createdAt: Date
+updatedAt: Date
EntryExitController
+checkIn(req, res)
+checkOut(req, res)
+openGate(req, res)
BillingService
+calculateFee() : : number
+calculateCycleFee() : : number
PricingPolicy
+userRole: enum
+vehicleType: enum
+calculationType: enum
+billingIntervalMin: number
+specialRules: array
+discountPercent?: number
+effectiveDate: Date
+createdAt: Date
+updatedAt: Date
User
+userId: string
+schoolCardId?: number
+fullName: string
+role: enum
+email: string
+userStatus: enum
+password: string
+createdAt: Date
+updatedAt: Date
+comparePassword()
Zone
+zoneId: string
+zoneName: string
+capacity: number
+currentUsage: number
+createdAt: Date
+updatedAt: Date
NavigationService
+getZonesByUsage()
+getStats() : : object
AuthService
+login(cardId, pwd) : : object
ReportService
+getDailyRevenue() : : array
+getActivityStats() : : object
SystemAdminService
+getAllConfigs()
+updateConfig()
+updatePricing()
+getPricingPolicies()
+getAuditLogs()
SystemConfig
+settingKey: string
+settingValue: object
+description?: string
+createdAt: Date
+updatedAt: Date
PaymentController
+initiateCyclePayment()
+mockBKPayCallback()
+getHistory(req, res)
+getHistoryAdmin()
+getDebt(req, res)
AuditLog
+userId: string
+action: string
+targetResource?: string
+details?: object
+timestamp: Date
+createdAt: Date
+updatedAt: Date
ParkingSession
+sessionId: string
+startTime: Date
+endTime?: Date
+sessionStatus: enum
+paymentStatus: enum
+type: enum
+userRole: string
+vehicleType: string
+subjectID: string
+plateNumber: string
+fee?: number
+invoiceId?: string
+createdAt: Date
+updatedAt: Date
IoTDevice
+deviceId: string
+locationId?: string
+status: enum
+lastOnline: Date
+zoneId: string
+deviceType: string
+deviceName?: string
+createdAt: Date
+updatedAt: Date
Location
+locationId: string
+locationName: string
+coordinates: array
+locationType: enum
+createdAt: Date
+updatedAt: Date
SessionController
+getAll(req, res)
+getByUser(req, res)
+getRecent(req, res)
ParkingSlot
+slotId: string
+zoneId: string
+isAvailable: boolean
+createdAt: Date
+updatedAt: Date
TemporaryCard
+tempCardID: string
+cardStatus: enum
+createdAt: Date
+updatedAt: Date
Invoice
+invoiceId: string
+amount: number
+paymentStatus: enum
+issueDate: Date
+createdAt: Date
+updatedAt: Date
Sensor
+parkingSlotId?: string
+sensitivity?: number
Gate
+gateType: enum
+ipAddress?: string
+isAutoOpen: boolean
Signage
+message: string
+brightness?: number
Camera
+streamURL: string
+resolution?: string
1
0..*
0..1

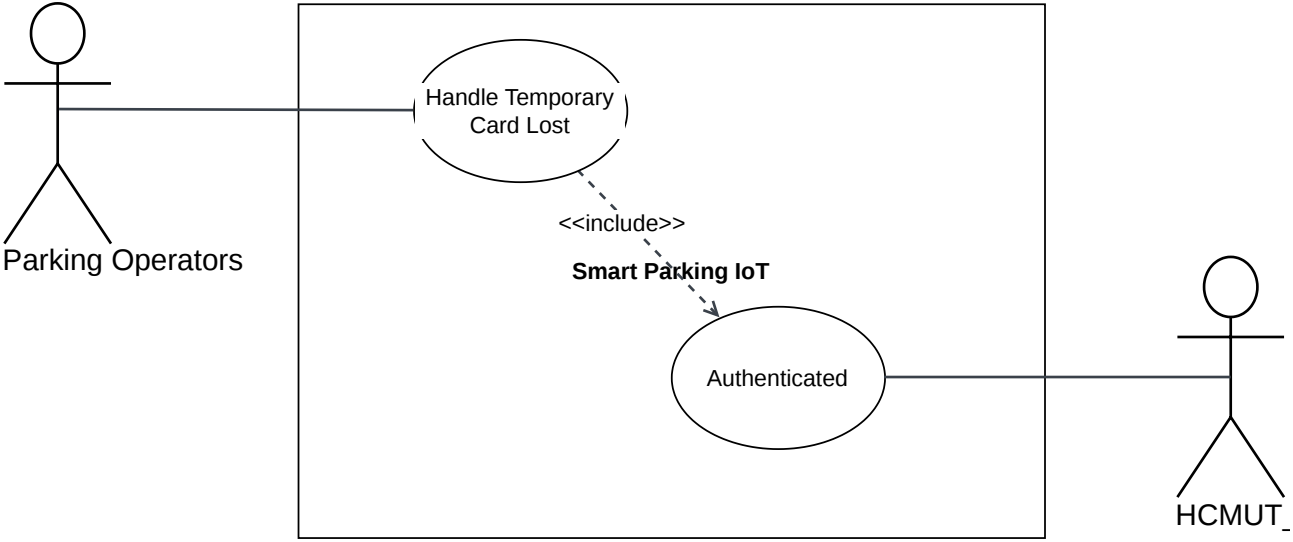

Learners
Faculty Members & Staff
Main Users
Enter Parking
<<include>>
Authenticate Parking Full State
Create Session
<<include>>
Check for unpay debt
Access Control - Entry Parking
<<extend>>
Manual Controlling
Only involves Gateway & Camera
Parking Operator
HCMUT_SSO
IoT Devices
Gateway
Sensors
Camera

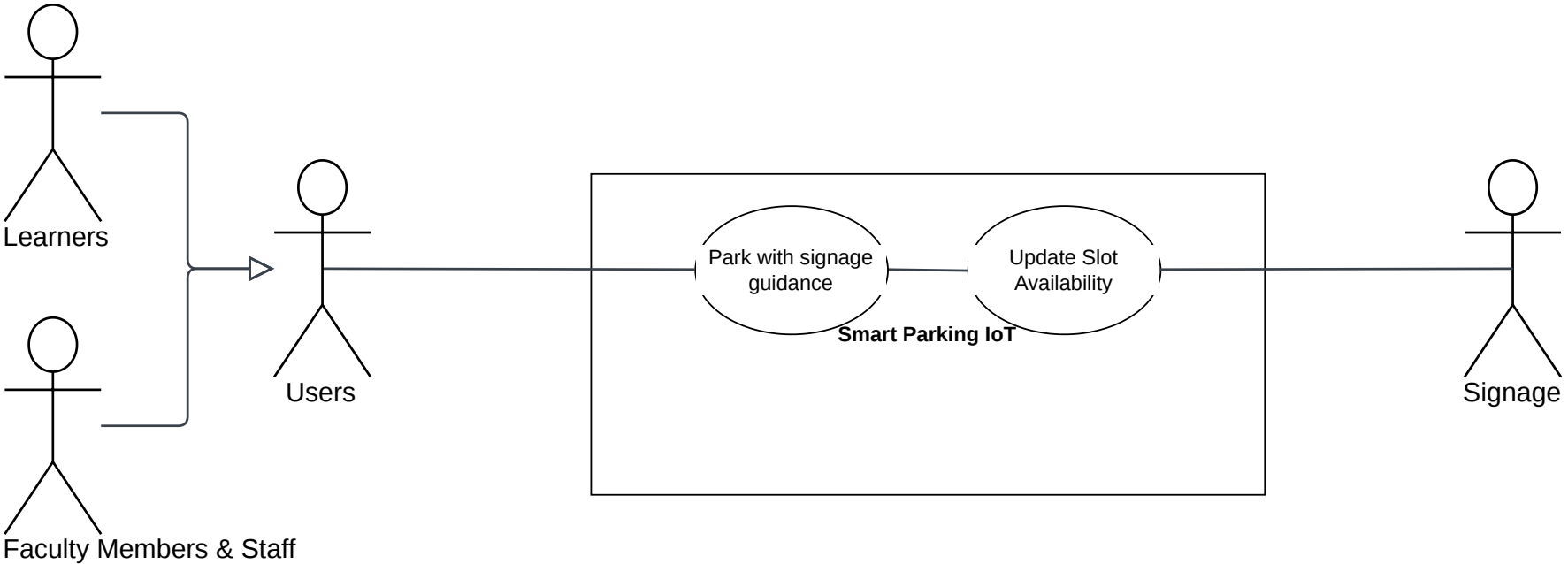

Create Session
<<include>>
Enter Parking (UC 1.2)
Temporary Visitors
Access Control - Entry Parking
<<extend>>
Manual Controlling
Only involves Gateway & Camera
Parking Operator
IoT Devices
Gateway
Sensors
Camera

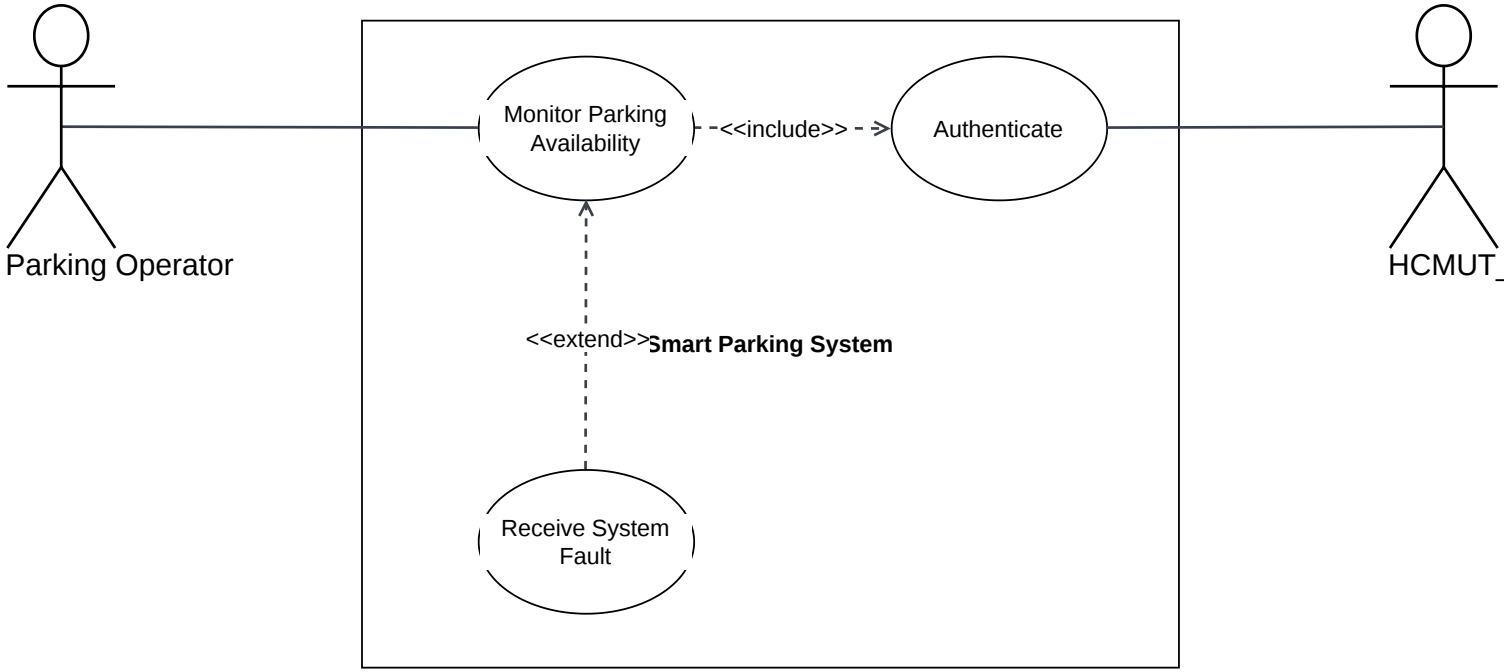

Learners
Faculty Members & Staff
Users
Exit Parking (UC 1.3)
<<include>>
Authenticate
<<include>>
Save session info
Access Control - Exit Parking
<<extend>>
Manual Controlling
HCMUT_
Parking Operators
Gateway

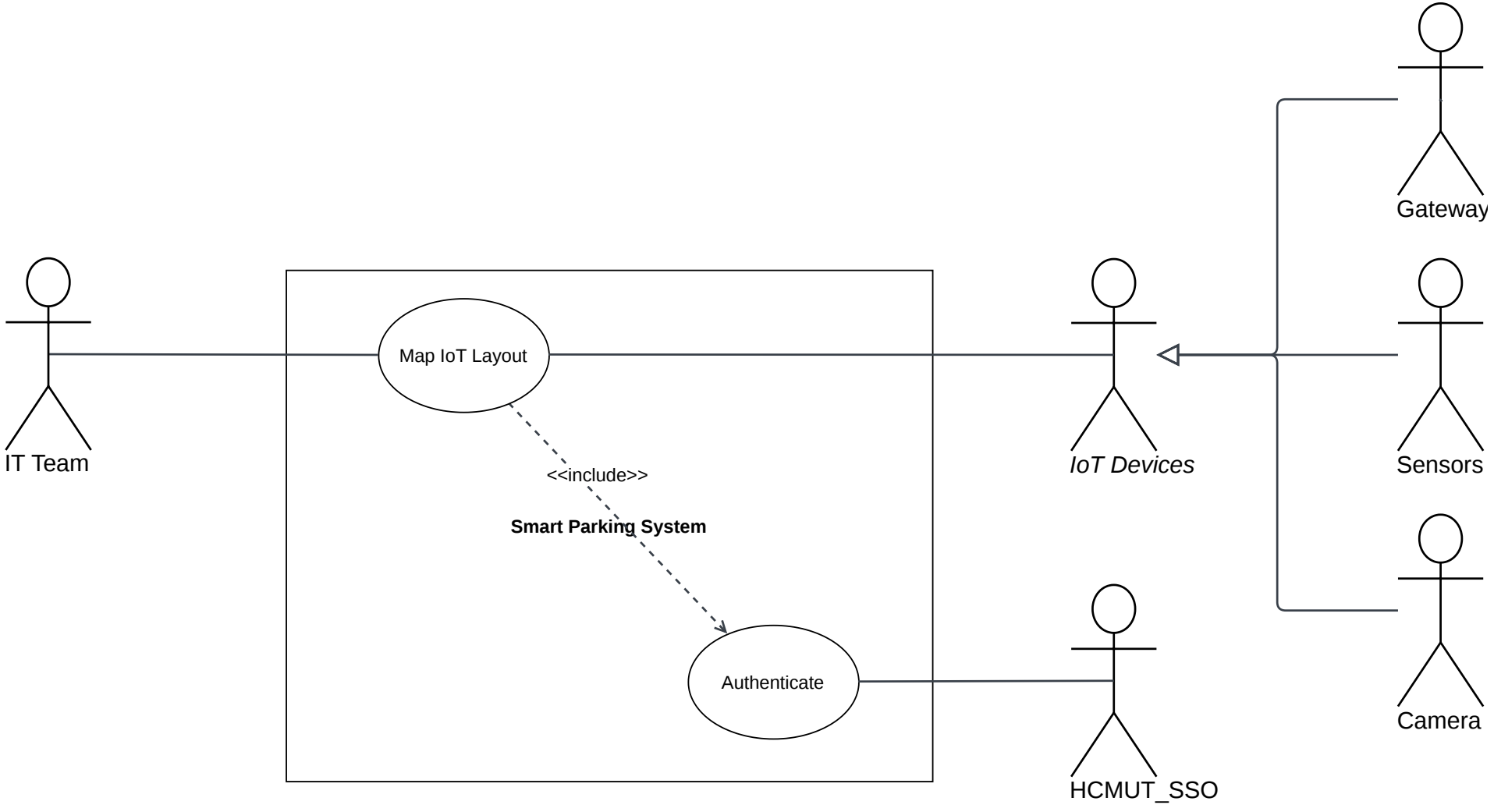

Parking Operators
Manual Controlling
Gateway
<<extend>>
Access Control - Exit Parking
Save session info
<<include>>
Exit Parking
<<include>>
Payment
Temporary Visitors
BKPAY

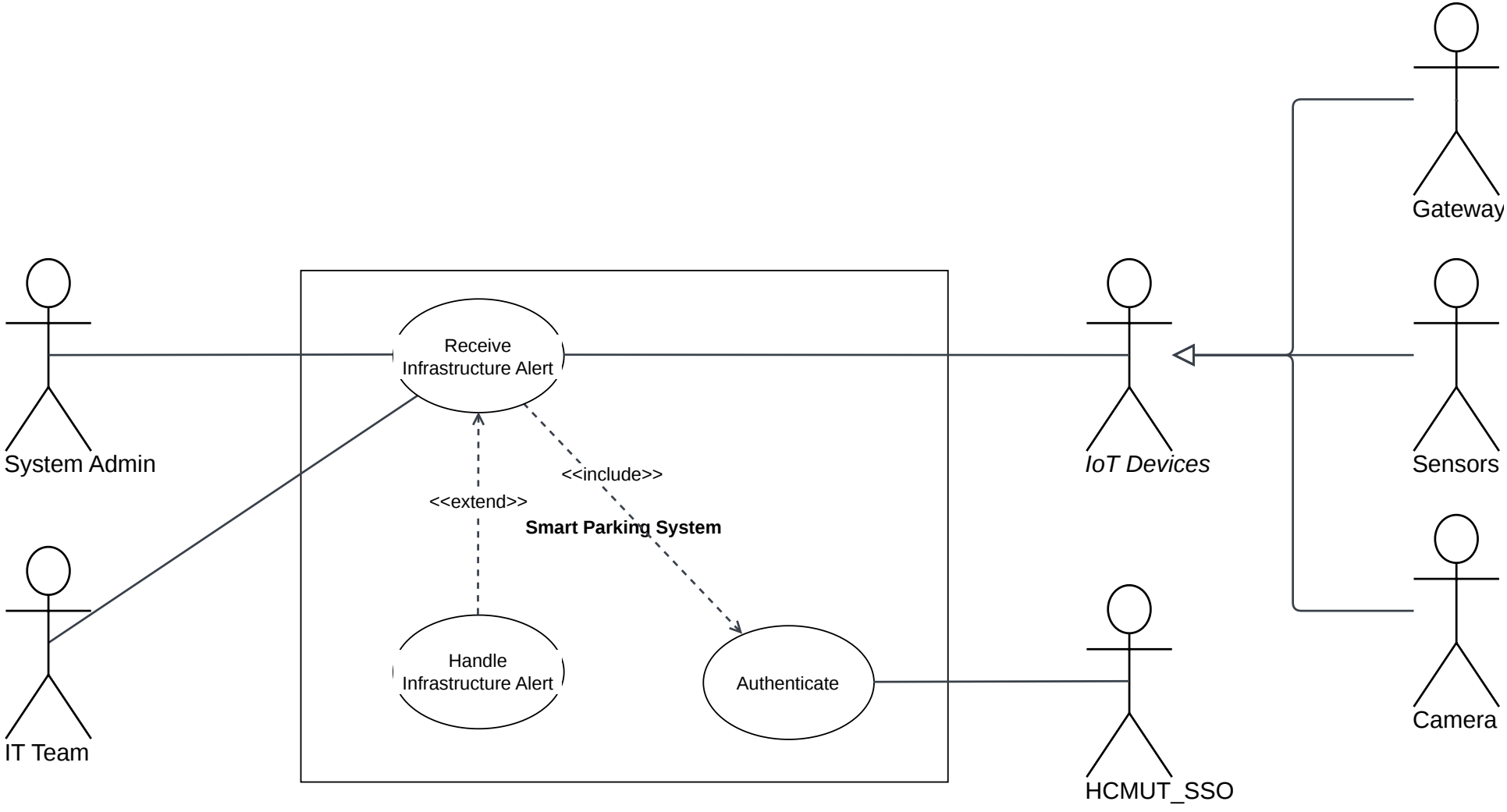

Parking Operators
Manually Gate Controlling
Audit Log
<<include>>
Smart Parking IoT
Authenticate
HCMUT_

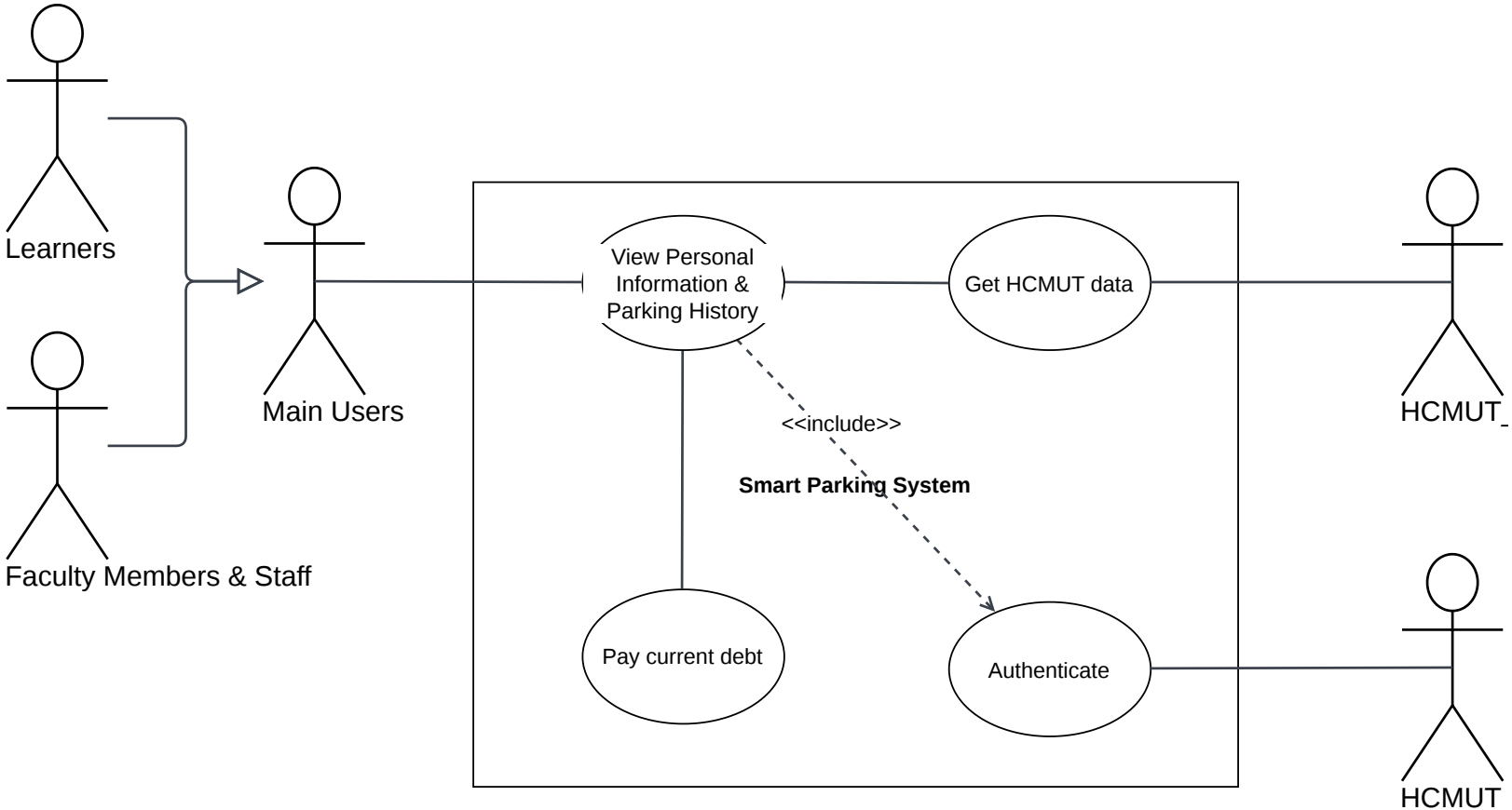

Parking Operators
Handle Temporary Card Lost
<<include>>
Smart Parking IoT
Authenticated
HCMUT_

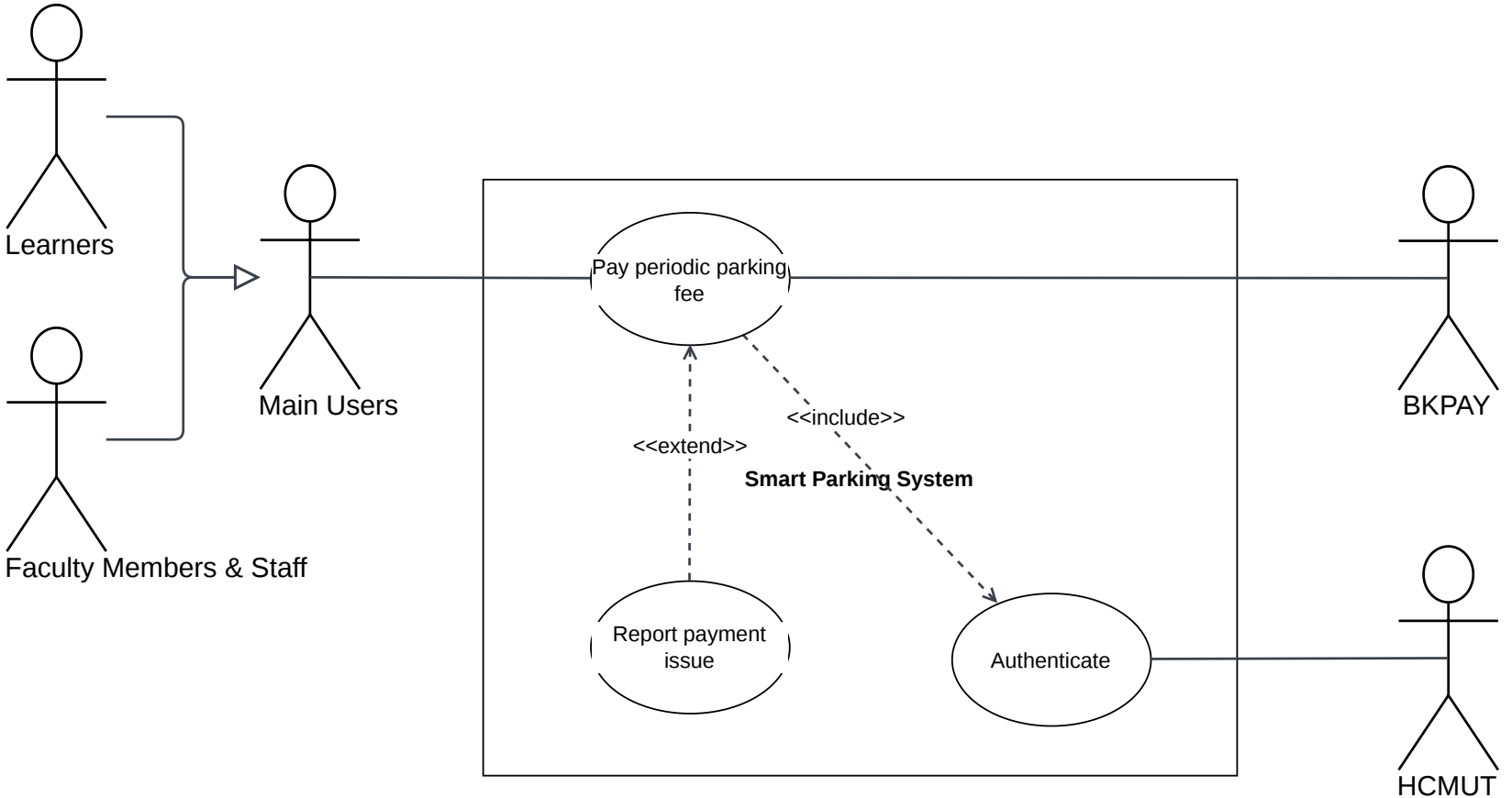

Learners
Faculty Members & Staff
Users
Park with signage guidance
Update Slot Availability
Smart Parking IoT
Signage

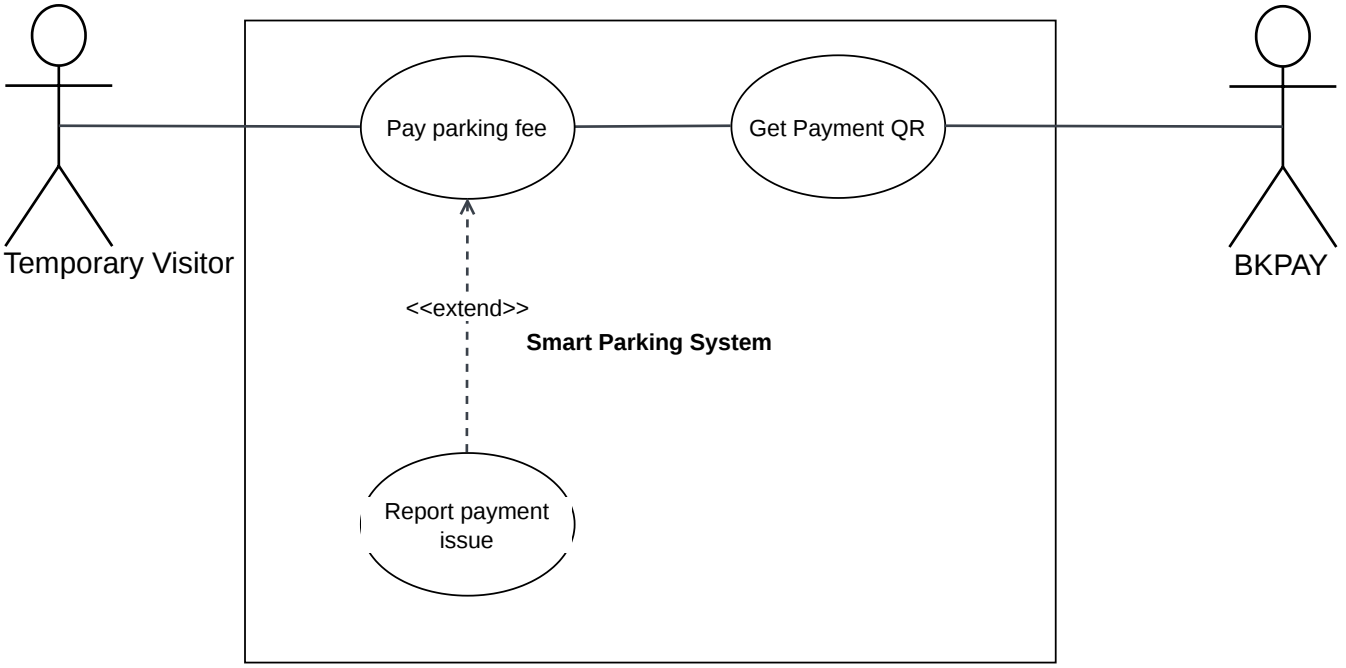

Parking Operator
Monitor Parking Availability
<<include>>
Authenticate
HCMUT_
<<extend>>
Smart Parking System
Receive System Fault

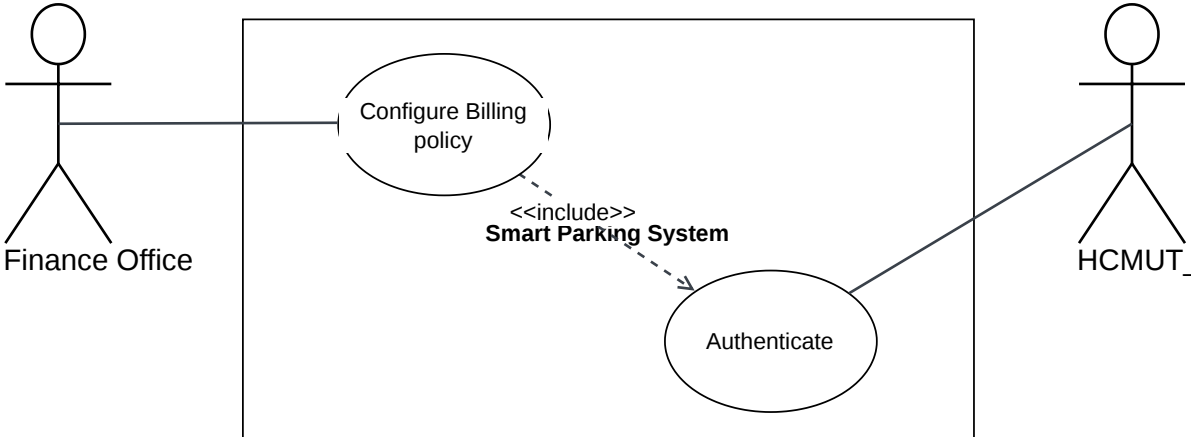

Gateway
IT Team
Map IoT Layout
IoT Devices
Sensors
<<include>>
Smart Parking System
Camera
Authenticate
HCMUT_SSO

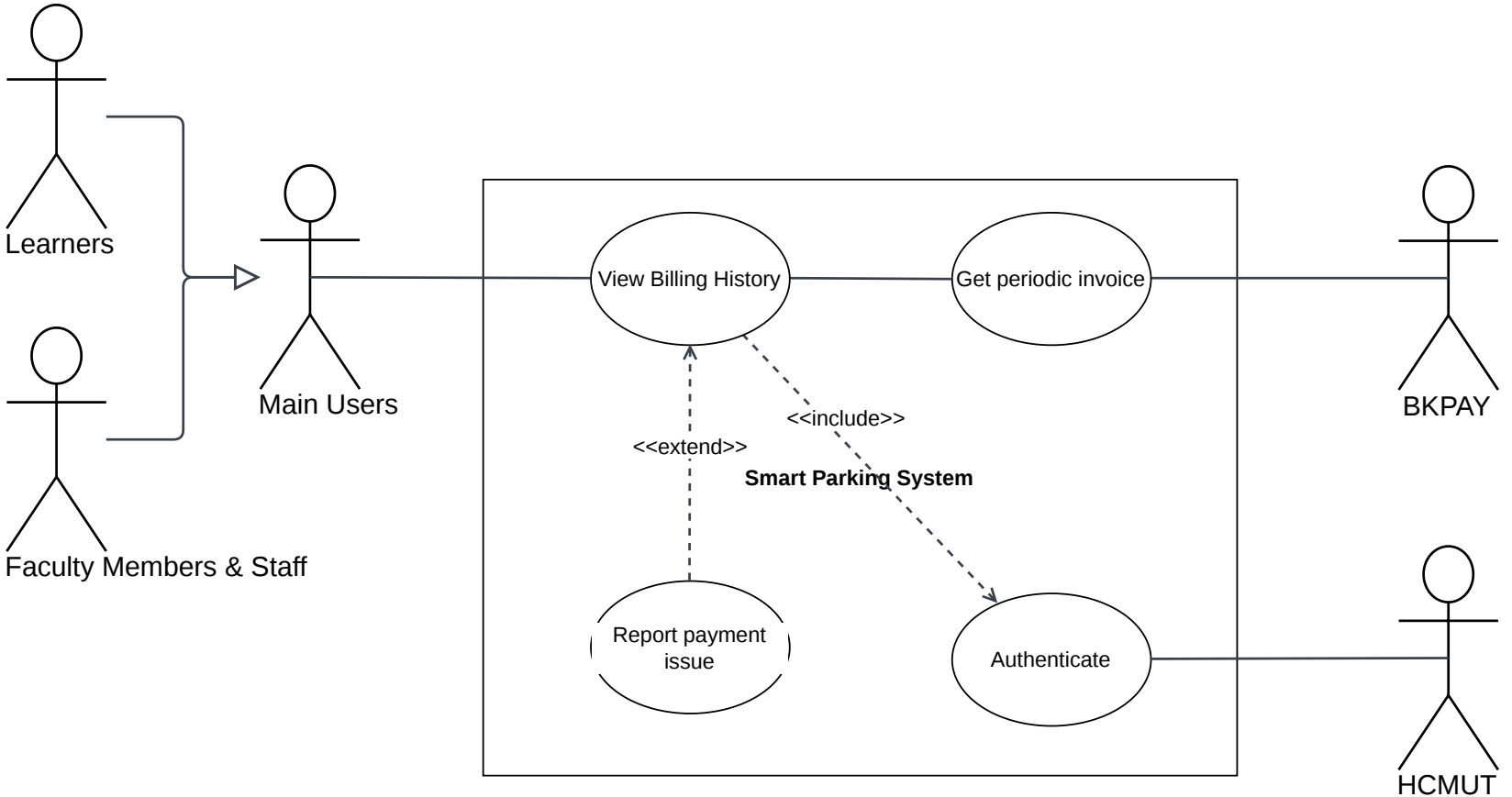

Gateway
System Admin
Receive
Infrastructure Alert
IoT Devices
Sensors
<<include>>
<<extend>>
Smart Parking System
IT Team
Handle
Infrastructure Alert
Authenticate
Camera
HCMUT_SSO

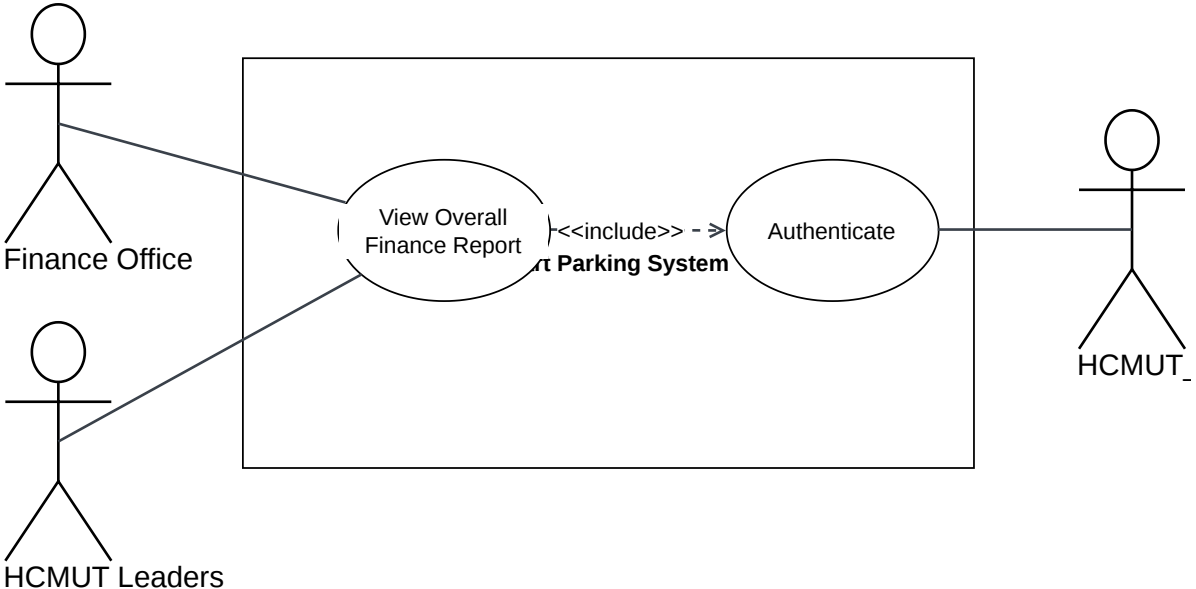

Learners
Faculty Members & Staff
Main Users
View Personal Information & Parking History
Get HCMUT data
<<include>>
Smart Parking System
Pay current debt
Authenticate
HCMUT_
HCMUT_

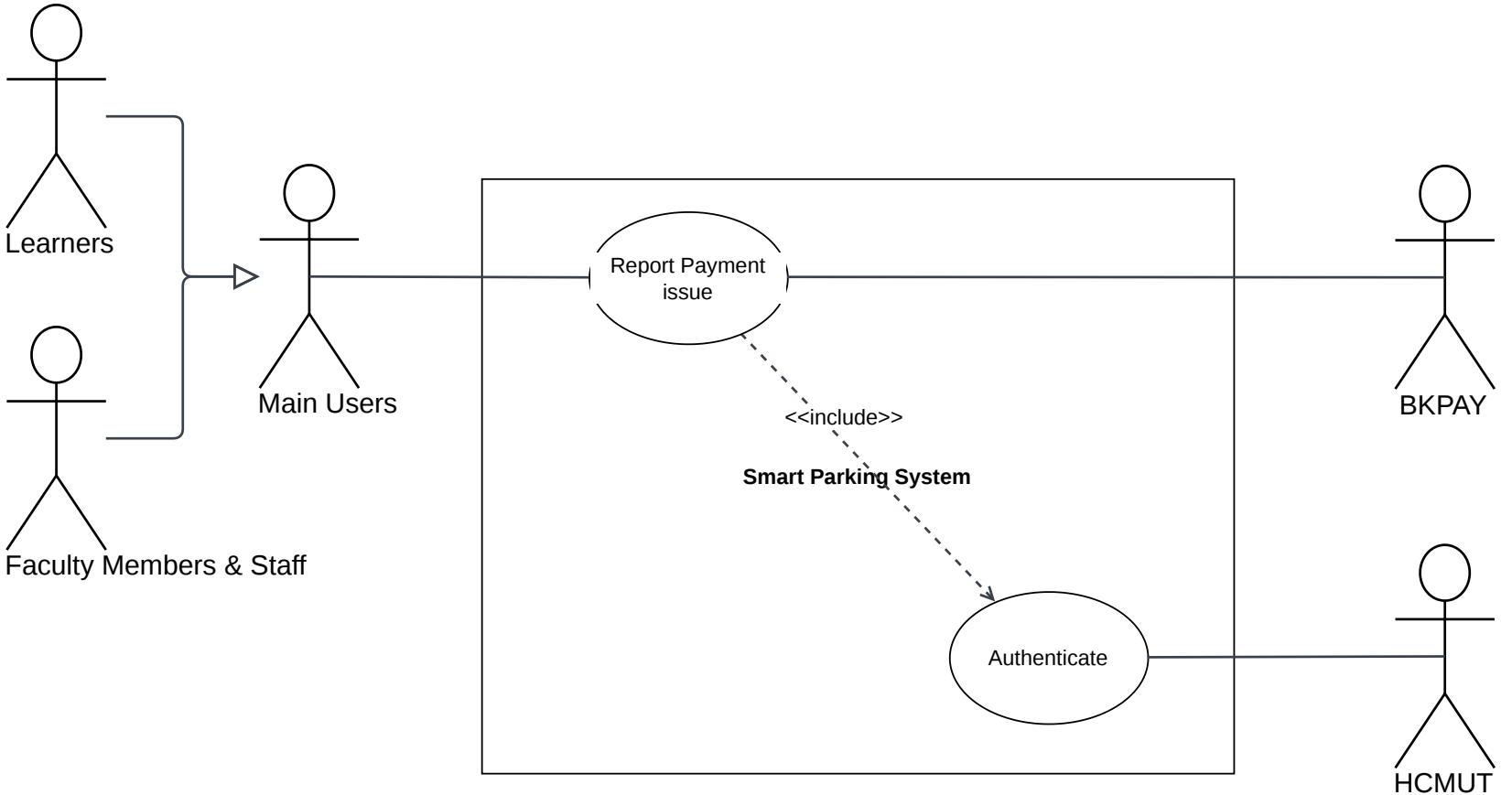

Learners
Faculty Members & Staff
Main Users
Pay periodic parking fee
<<include>>
<<extend>>
Smart Parking System
Report payment issue
Authenticate
BKPAY
HCMUT_

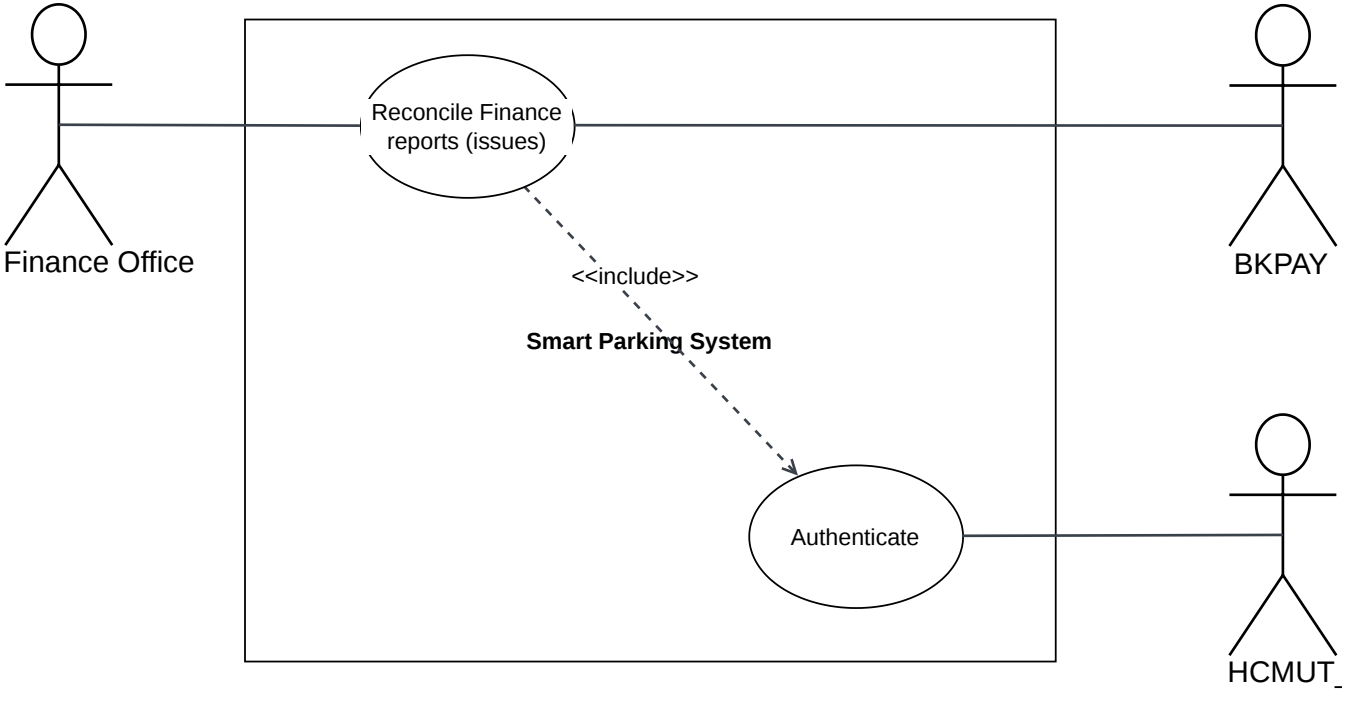

Temporary Visitor
Pay parking fee
Get Payment QR
BKPAY
<<extend>>
Smart Parking System
Report payment issue

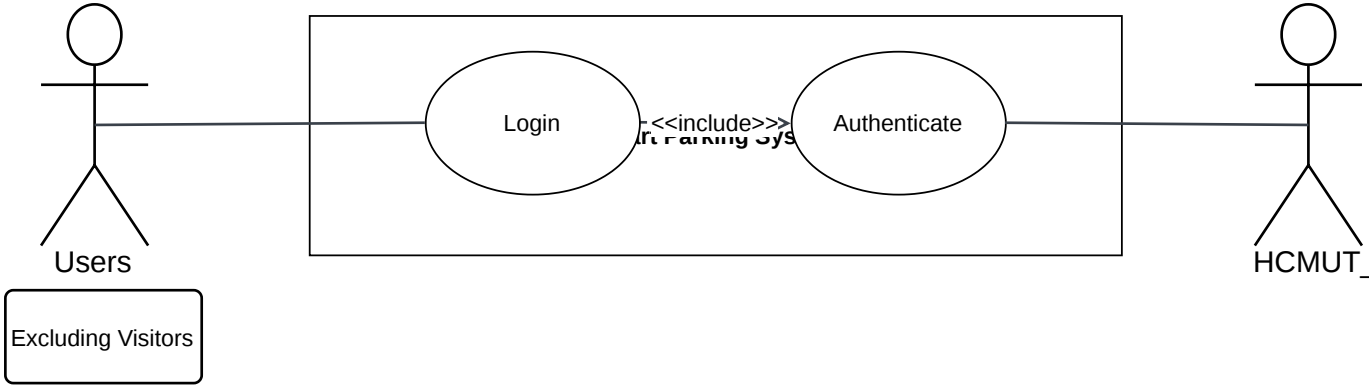

Finance Office
Configure Billing policy
<<include>>
Smart Parking System
Authenticate
HCMUT_

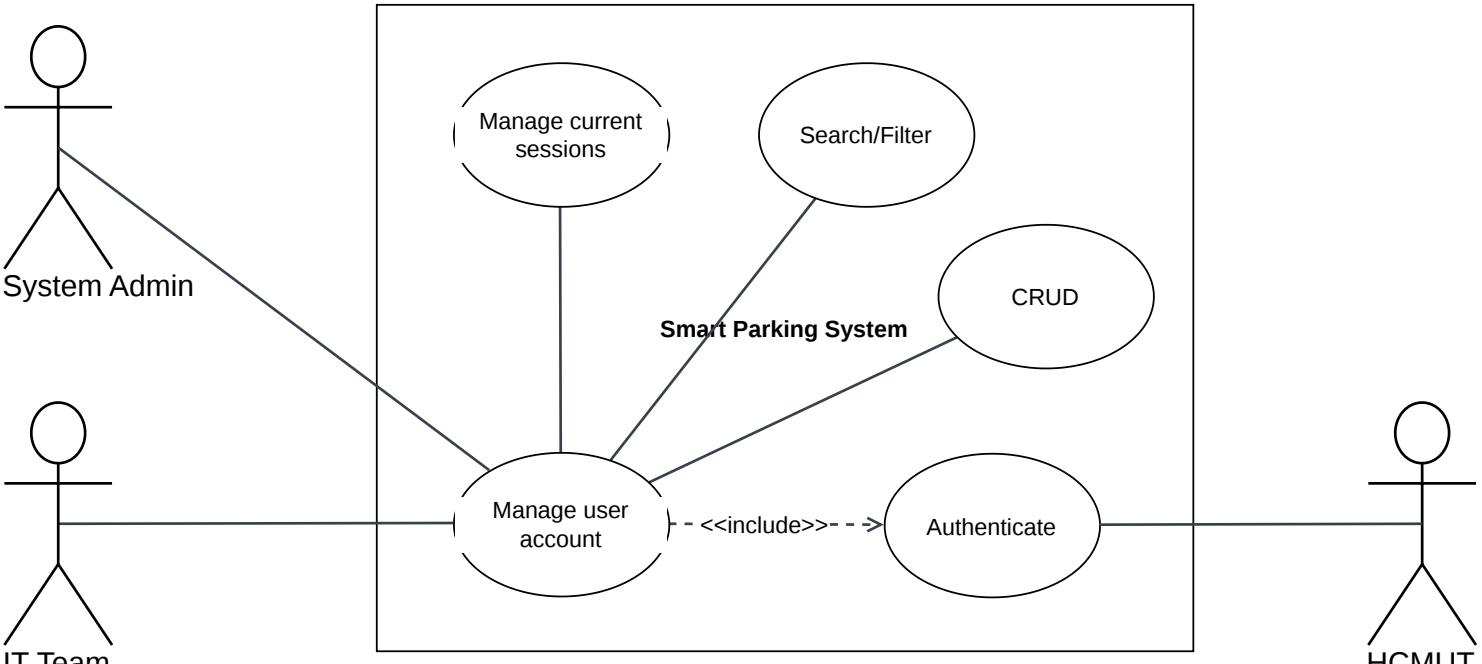

Learners
Faculty Members & Staff
Main Users
View Billing History
Get periodic invoice
<<include>>
<<extend>>
Smart Parking System
Report payment issue
Authenticate
BKPAY
HCMUT_

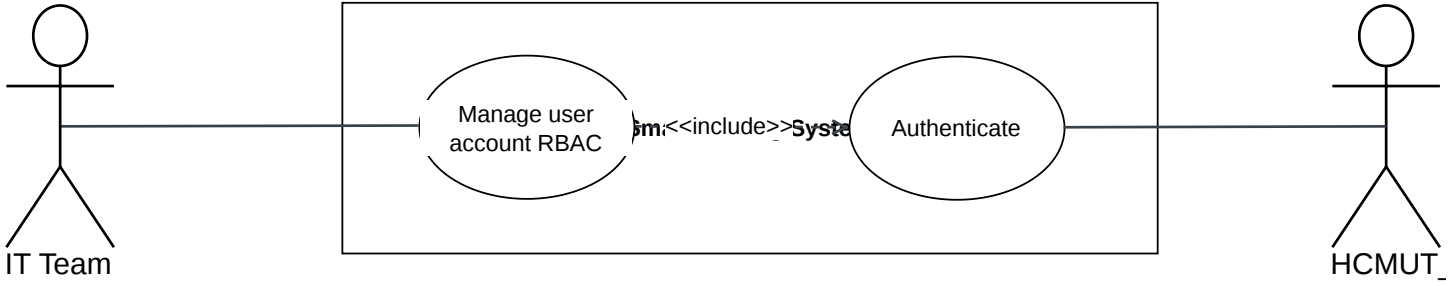

Finance Office
View Overall
Finance Report
<<include>>
Authenticate
rt Parking System
HCMUT_
HCMUT Leaders

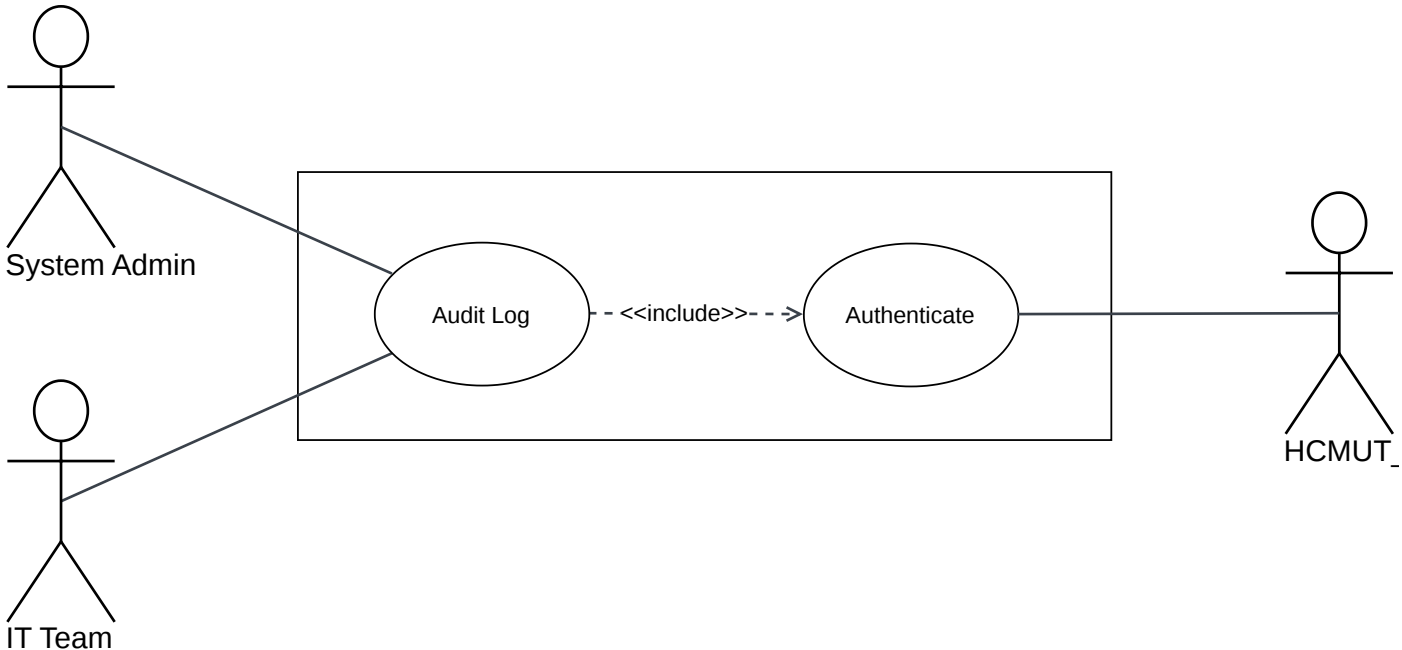

Learners
Faculty Members & Staff
Main Users
Report Payment issue
<<include>>
Smart Parking System
Authenticate
BKPAY
HCMUT_

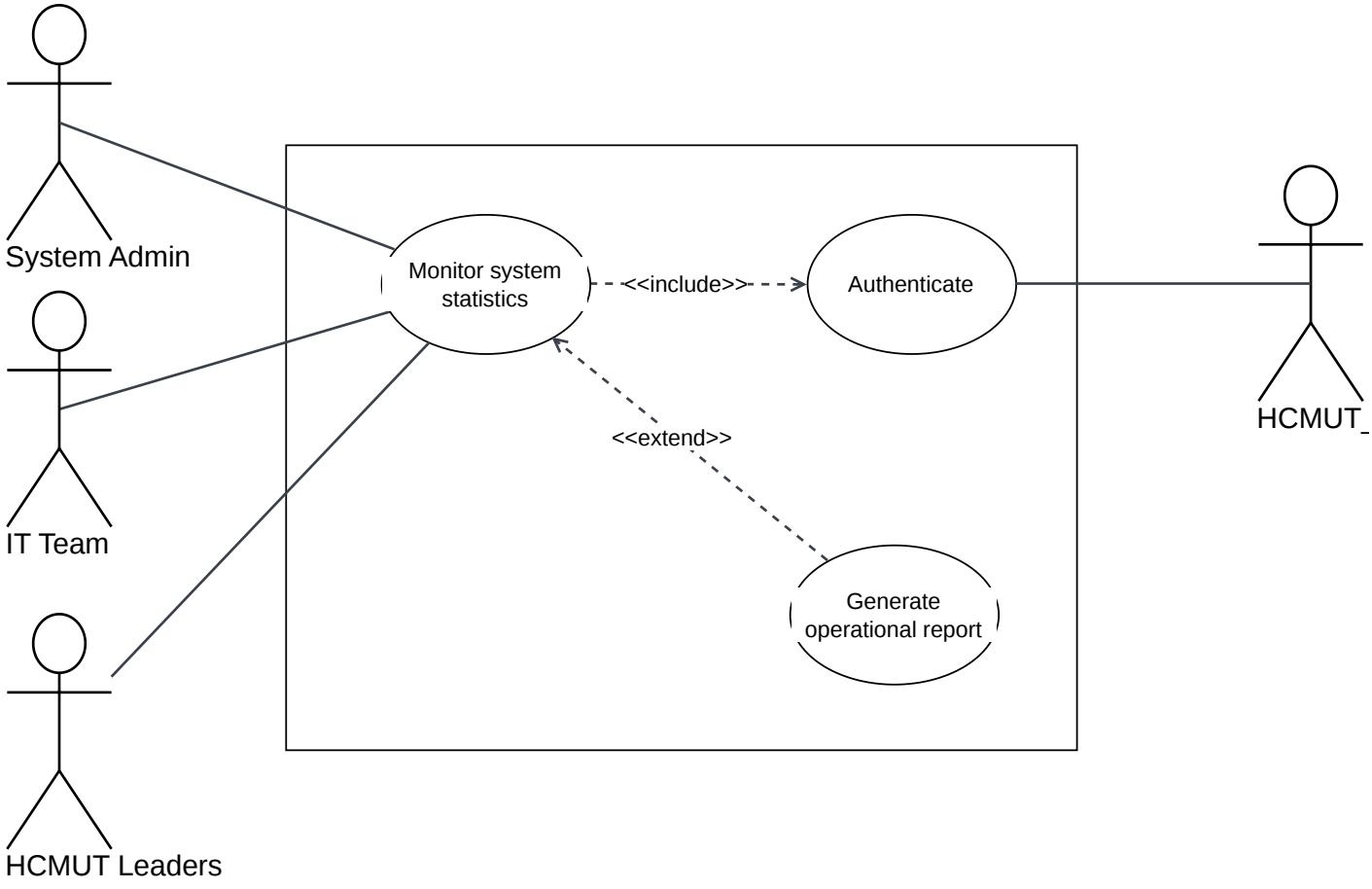

Finance Office
Reconcile Finance reports (issues)
BKPAY
<<include>>
Smart Parking System
Authenticate
HCMUT_

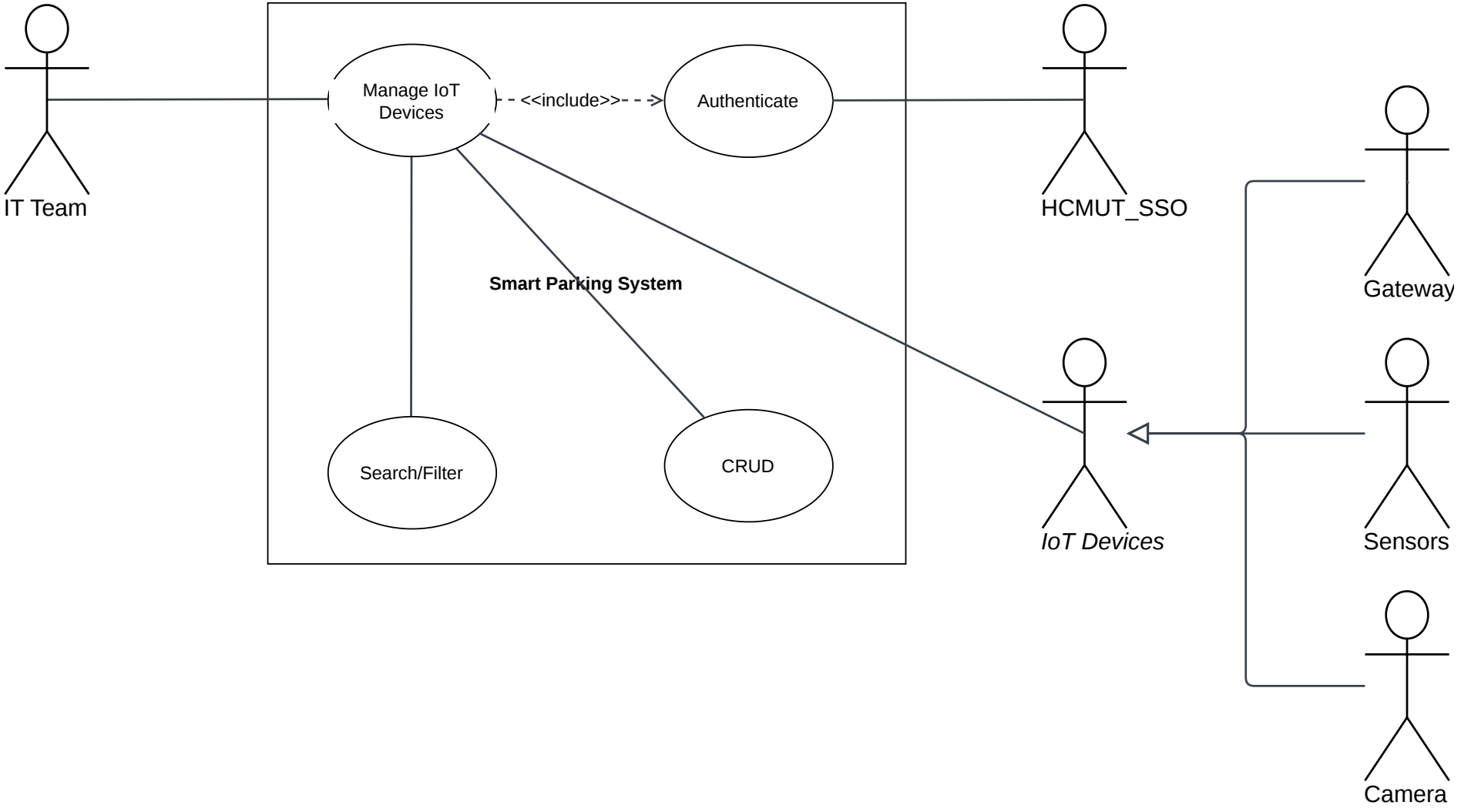

Users
Login
<<include>>>
Authenticate
HCMUT_
Excluding Visitors

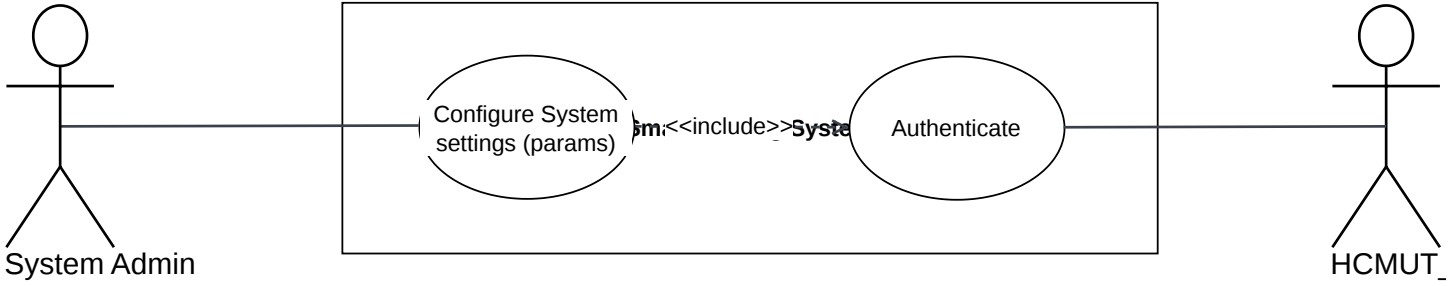

System Admin
IT Team
HCMUT_
Smart Parking System
Manage current sessions
Search/Filter
CRUD
Manage user account
<<include>>
Authenticate


IT Team
Manage user account RBAC
<<include>>
Authenticate
HCMUT_


System Admin
IT Team
Audit Log
<<include>>
Authenticate
HCMUT_


System Admin
IT Team
HCMUT Leaders
Monitor system statistics
<<include>>
Authenticate
<<extend>>
Generate operational report
HCMUT_


IT Team
Manage IoT Devices
<<include>>
Authenticate
HCMUT_SSO
Smart Parking System
Search/Filter
CRUD
IoT Devices
Gateway
Sensors
Camera


System Admin
Configure System settings (params)
<<include>>
Authenticate
HCMUT_